2026 2 — 2015·2022

3

/ (· ·) | 45 | 25 (20, 5) | 100 | , ,

—

- · , .
- **5 1** . .
- **( )** .
- , .

( )
[9 13-01~04] · [9 14-01~05] · · [9 15-01~04]
· 40% / · 35% / · 25% · 30%
50% 20%
. . , ,

· ·DNA· · · - · · ·HR · · · · · · · ·

4 ×20=80 , 4 ×5=20

. 1 ~8 —

1. [·] — [4] [9 13-01]

— ?

.

.

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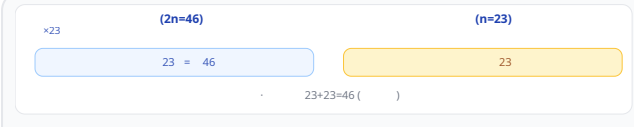
.

.

2. [ ] — [4] [9 13-01]

. (, n=23) .

2n=46, n=23 —



[ ] — ( )

( ) .

.

.

.

23 .

3. [·] DNA- - — [4] [9 13-02]

DNA, , < > .

< >		
.	DNA .	
.	, DNA .	
.	.	

,

,

,

, ,

4. [·] — [4] [9 13-02]

(R) (r) , (RR) (rr)
1 (F_1, Rr) . 2 (F_2) ,
(RR) ?

(Punnett square) .

F_1(Rr) × F_1(Rr)

	R	r	
R	RR	Rr	: **= 3:1**
r	Rr	rr	

—

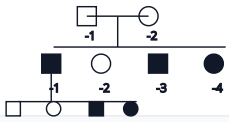
[ ] F_1 — 3:1

1/4, 1/3
1/2, 1/2
3/4, 1/3
3/4, 2/3
1, 1/4

5. [ ] — — [4] [9 13-03]

. .
?
: · , .

(: , :)



-1( )× -2( ) -1( )
:
-1( )× ( ) .

[ ] — (-1, -2) (-1)

, -1 .
, -1 -2 (heterozygous) .
, .
, .
. .

6. [·] — [4] [9 13-03]

?

X Y .
X Y .
.
X .
.

7. [ ] vs — [4] [9 13-04]

< > .

< >		
.	(/)	
.	( )	
.	ABO	
.		

,
,
,
,
,

8. [ ] — [4] [9 13-04]

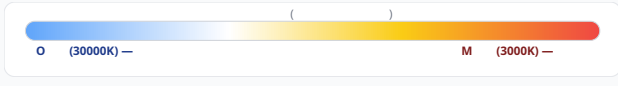
, .
.
: 2 + 2 (1 ,)

(8)

: (1) ___ ___ . (2) : ___

9. [·] — [4] [9 14-01]

?



[ ] OBAFGKM —

.

.

.

.

.

10. [ ] — [4] [9 14-02]

A, B . .

	(m)	(M)
A	1.0	-5.0
B	1.0	2.0
: 10pc /		

A B .

( ) .

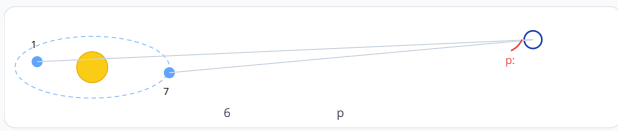
A B .( )

.

.

11. [ ] — [4] [9 14-02]

(p) 0.1″ pc ?(, (pc) = 1 / (″))



[ ] —

0.1 pc

1 pc

10 pc

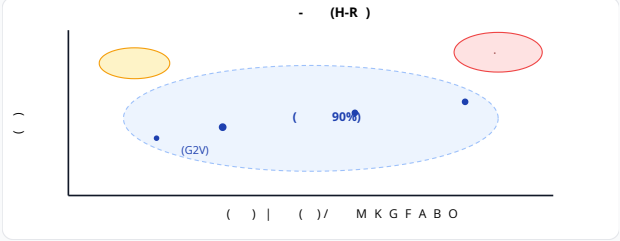
100 pc

1000 pc

12. [ ] HR (-) — [4] [9 14-03]

HR ( ) . HR

.



[ ] HR — (·) (·)

, 90% .

HR .

.

.

HR .

13. [·] — [4] [9 14-04]

?

14. [ ] — [4] [9 14-04]

?(, = 1M)

.

10M 100 .

.

100 .

.

15. [ ] — [4] [9 14-05]

?

.

.

.

.

.

16. [ ] — [4] [9 14-05]

(:) , ‘ ’

2 .

(16)

: / 3 /

-2. 17 ~25 —

17. [·] — [4] [9 15-01]

— ?

.

.

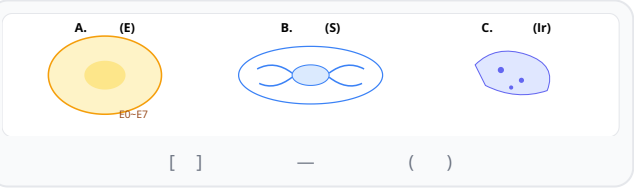
( ) .

2000 .

.

18. [ ] — [4] [9 15-01]

. A~C .



[ ] — ( )

A .

B .

C .

.

C .

19. [·] — [4] [9 15-01]

?

.

,

.

.

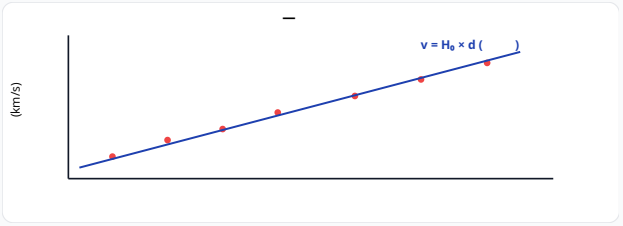
.

.

20. [ ] — [4] [9 15-02]

( ) .

?



[ ] ( ) — (SDSS) ( )

.

— .

.

.

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21. [·] — [4] [9 15-03]

(H_0) 70 km/s/Mpc , 100 Mpc km/s ? (, $v = H_0 \times d$)

	$v = H_0 \times d$
	1 Mpc = 326

0.7 km/s

7 km/s

70 km/s

700 km/s

7000 km/s

22. [·] — [4] [9 15-04]

< > .

< >	
.	( )
.	(2.7K)
.	

,

,

,

, ,

23. [ ] · — [4] [9 15-04]

? (' ')

(95%) .

.

.

.

.

24. [ ] — [4] [9 15-02~04]

3 .

: · · / ·

(24)

25. [·] — [4] · ( )

" " .

. ,' . .

' 2~3 .

(25 —)

: · · 2 (2) + / (2)